Practical 1

* Aim :- Intrduction SQL and Oracle Installation.

* Theory:- SQL (Structured Query Language) is a standard programming language used to manage and manipulate relational databases. It allows users to perform various operations such as creating databases structures, inserting data, updating records, deleting data, and retrieving information efficiently. SQL is widely used because of its simplicity, flexibility, and powerful querying capabilities.

SQL commands are categorized into different types :-

① DDL (Data Definition Language) :- Used to define databases structure (CREATE, ALTER, DROP)
② DML (Data Manipulation Language) :- Used to modify data (INSERT, UPDATE, DELETE)
③ DQL (Data Query Language) :- Used to retrieve data (SELECT).
④ DCL (Data Control Language) :- Used for access control (GRANT, REVOKE).
⑤ TCL (Transaction Control Language) :- Used to manage transactions (Commit, ROLLBACK)
SQL works with relational databases

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where data is stored in tables consisting of rows and columns. It supports constraints such as Primary key, Foreign key, unique key and Not Null to ensure data integrity.

MySQL is an open source relational database management system that uses SQL to manage data. It is widely used in web applications and supports multi-user access and high performance.

MySQL Workbench is a graphical user interface (GUI) tool used to interact MySQL databases. It allows user to write SQL queries, design databases visually, and manage databases operations easily without using command-line tools.

Step 1:- Download MySQL Installer
* Go to the official MySQL website
* Download MySQL Installer for Windows
* Choose the "Community Version".

Step 2:- Run the Installer
* Open the downloaded .exe file
* The setup wizard will start automatically.

Step 3:- Choose setup type
You will see different options:-
* Developer Default (Recommended) → Installs

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MySQL Server + Workbench + other tools
* Server Only → Only installs database server
* Custom → Choose Components manually
Select Developer Default for lab work

Step 4:- Install Required Components
* Click Next
* The installer will check for missing requirement.
* If anything is missing, click Execute to install them
* Then click Next

Step 5:- Configure MySQL Server
* Choose standalone MySQL Server
* Keep default settings
* Set a root password & remember it.

Step 6:- Configure Windows Service
* Keep default service name
* Select Start MySQL Server automatically
* Click next.

Step 7 :- Apply Configuration
* Click execute to apply settings
* Wait until all steps show Completed
* Click Finish

Step 8:- Launch MySQL Workbench
* Open MySQL Workbench from Start Menu
* Click on Local Instance MySQL
* Enter the root password

Step 9:- Test the installation
* Open a new SQL tab
* Run:-

```
SHOW DATABASES;
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If works, installation is successful

* Conclusion:- In this practical, we learned the basics of SQL and how to install the MySQL Workbench. It helped to understand how databases are created & managed using a graphical interface, making database operations easier and more efficient.